

Balancing Power Relations in Participatory Design: The Importance of Initiative and External Factors

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ABSTRACT

Power imbalances between users and designers impede the intent to equal contributions in a participatory process. Recent years have shown that unforeseen external factors pose a risk of exacerbating this power imbalance by limiting in-person meetings and communication in general. This study evaluated a projects' power relations using a three-part reflection-on-action approach. Results show, that external factors can act as an actor in the power relationship model and that the change of initiative can change the power relations. Thus, we propose a power relation triangle for Participatory Design processes, including participants, designers and external conditions as actors and the decision-making at the center based on the combination of the actor's relations. This framework can help to better understand and address power imbalances in Participatory Design.

CCS CONCEPTS

• **Human-centered computing** → *HCI theory, concepts and models*; **Empirical studies in collaborative and social computing**.

KEYWORDS

participatory design, older adults, gerontechnology, user-collaboration, power, decision-making

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1 INTRODUCTION

Participatory Design (PD) is a collaborative process to co-design with users the technologies they use [24]. It has the potential to involve marginalized user groups, such as older adults, more in the development process, and thereby to better address their needs and wishes [15, 21]. Typically, in technological development processes there is a power imbalance between those who develop the technology, namely designers (i.e. researchers and developers) and the later users [9, 20]. This gap also persists in participatory processes,

e.g. due to different levels of technological expertise. Moreover, recent years have shown that unforeseen external factors, such as the COVID-19 pandemic, pose a risk of exacerbating this power imbalance by limiting in-person meetings and communication in general. In the development of socio-technical systems, it is important to understand all the actors and factors involved, as well as their power-relational interplay. An important question in this context might be who initiates a participatory process or collaboration. Taking the initiative may be a way for certain actors to balance power relations to their benefit. One such example is described in the following, where participants themselves approached the designers and took on the role of initiator of the participatory design process. In this deviation from the human centered design process, we address the question of what is changing in terms of power relations and how important the role of initiative is in PD.

2 RELATED WORK

Participatory Design (PD) originated in the political landscapes of Scandinavia and North America in the 1960s and 1970s, aiming to give workers a voice in the development of their own workplace and technological progress. Based on these projects, a research community with its own objectives and values evolved in the 1990s [13, 25, 29]. Since then, PD has become increasingly important in the development of technical solutions and has been able to establish itself especially in projects with social and economic challenges [1]. If done right, PD can be used to reveal the needs especially of marginalized user groups [3, 7, 10, 18, 32].

Basic principles of PD are (1) design for empowerment and thus involvement of the users in all development phases from analysis to evaluation [18], (2) democratization of the design process [16, 25, 32], (3) building trust amongst the participants and facilitate mutual learning among all participants [13], (4) inclusion and cooperation [13, 19], and (5) quality of life improvements through advancements of technology [13]. Thus, PD is a process of researching, understanding, and reflecting mutual learning among different participants in a collective *reflection-in-action* [23]. It changes the notion of designers, developers, and/or researchers who act as a moderator to empower the user group and less as the designer themselves. Users are seen as experts in their own environment. In a collaborative design, their knowledge is as important as the knowledge of the designers, developers, and research. PD should facilitate an iterative collaboration between all stakeholders within the whole development process. [2, 22, 26].

2.1 Power Relations and Participatory Design

Analyzing power and locus of control has always been a core concern of Participatory Design research [15]. Thus, one core principle

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is giving the users *power over* their own environment and the design of the digital systems they use [8]. The execution of power is decision-making and decision-making has high significance to design digital systems [4, 8].

The focus of attention in the complex interplay of power sharing is mainly fixed on the designers and users. Only in the last few years, non-human actors are increasingly considered as influencing factors on participatory design results [14]. Over the years, a design ethos has been established that designers have the right expertise to wield *power over* the users of a system, who then exercise decision-making authority within the process [4, 11, 27]. The influence of the designer within a collaborated process is shaped by

- trust and mutual understanding [4],
- their skills, motivations, practices, relationships, and their priorities [15],
- the understanding of cultural practices and local power relations [33].

Furthermore, they typically organize, plan and select methods, tools and techniques as well as wield the power of *making*, which brings them into a powerful position within system design [6]. PD has the values, methods and tools to oppose this ethos in an aim to conjoin the voices of designers and users to embrace discussions and collective decision-making [4, 27]. Thus, PD literature suggests that designers should reduce their “status as authority figures” [33] and act as “facilitators” [27], “gamemasters” [14] or should be viewed “as [just] another stakeholder in the process” [17] instead.

PD literature suggests that users on the other hand can contribute through their own language and skill set in creating choices, selecting choices, evaluate systems and drive further development from defining the problems to delivering ideas [6]. Their power is influenced by context knowledge and thus knowing the system in use [8] and various factors that are difficult to grasp and must be explored for an individual context [15].

The third component that influences the power dynamics of the interplay of users and designers are external factors. They consist of resources (i.e., time, money, and assistance) [14], the material [14], unexpected events [12], and dynamic or new local contexts [12, 15, 33].

From the previously described research findings, a preliminary power-relational model for PD processes can be derived, which is shown in figure 1. It consists of a bidirectional relationship between the main actors of the PD process, namely designers and participants.

3 METHOD

The participatory process is performed within a project called Historytelling (HT) [30]. It aims to empower older adults to record and share their life stories on a social digital platform as well as to take an active part in the development of said platform. HT impacts sociality on (1) personal level, offering possibilities of reminiscence and biography work; (2) family level, enabling cross-generational transfer of knowledge as well as the formation of new groups or friendships; and (3) societal level, allowing users to experience history in a multi-perspective and multi-media format, appreciating older people’s experiential knowledge.

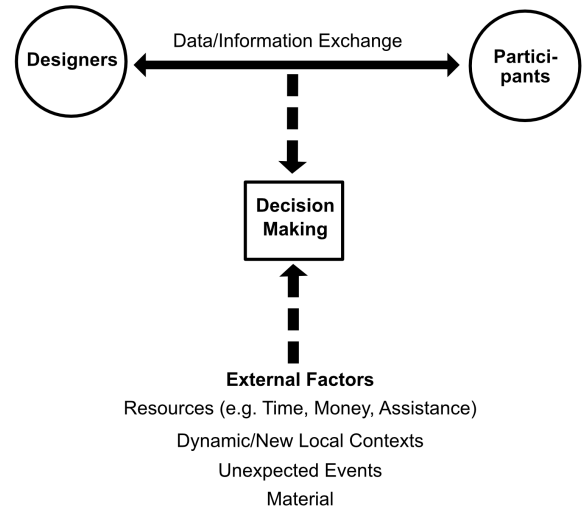


Figure 1: Literature-based Model of power-relations in participatory processes

Representing one of the key target groups for HT, a local chapter of the RuralWomen Association¹ has been extensively involved in the participatory development for many years. Thus, a relationship of trust between them and those involved in HT research and implementation has developed [31]. However, this relationship as well as the further development of HT was confronted with sudden unforeseen challenges, when the COVID-19 pandemic caused in-person meetings to be cancelled for an unbeknown time. On the initiative of the RuralWomen, the idea of using HT as an alternative solution for cancelled or postponed events, that could still take place through digital formats, arose.

3.1 Participants

The participants of this participatory process can be roughly divided into two categories: (1) members of the RuralWomen (German: “LandFrauenVerein e.V.”) – predominantly older adults ($N = 9$ to 13) as well as (2) designers, who consist of students and research assistants ($N = 5$).

3.2 Process

At the beginning of the project, we conducted an online planning meeting to discuss the goals, needed roles and requirements for collaboration. The meeting was followed by five consecutive online workshops, and additional informal phone calls between participants and designers on the video conferencing solution Cisco Webex². Webex was chosen because it was already familiar to participants from past events. In addition, the collaborative whiteboard platform Miro³ was used to visualize and discuss ideas. Communication regarding workshops invitations, the topics and intermediate

¹<https://www.landfrauen.info/>

²<https://www.webex.com/>

³<https://miro.com>

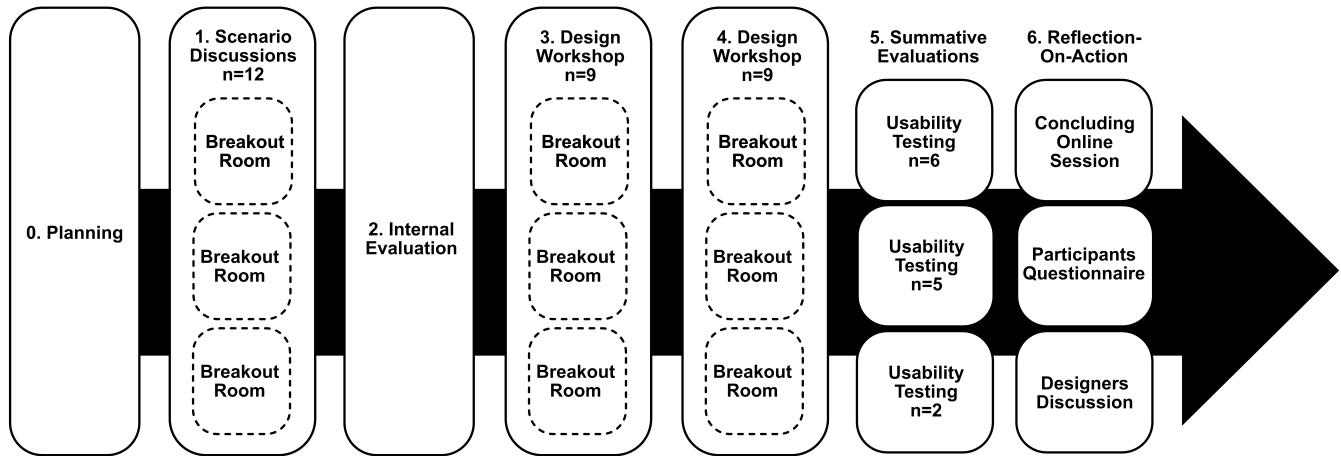


Figure 2: Depiction of the participatory process consisting of several online meetings: 0 planning meeting, 1. a scenario discussion workshop (3 split meetings), 2. an internal evaluation (designers only), 3.4. two design workshops (3 split meetings), 5. a summative evaluation of the developed system, and 6. evaluation of the process via the reflection-on-action approach

results was handled via WhatsApp⁴. This was the preferred communication channel of the members of the RuralWomen association. They set up a chat group for the project in which all interested members of the association could participate. At the request of the older adults, the workshops were scheduled for 90 minutes on Saturday afternoons due to other commitments and fear of screen fatigue. In reality they lasted 90-120 minutes. In the five workshops, the participant number varied between 9 and 13 members of the RuralWomen association and 3 and 5 designers. The current status of the project was discussed with everyone together at the beginning and new insights from the participants and their intermediate work were gathered. Work was done in separate breakout rooms leading into a plenary discussion to compile results. See Figure 2 for an overview of our participatory process.

In the end, we conducted a three-part evaluation as reflection-on-action approach [5]. First, a qualitative analysis of the implemented methodology was conducted during the concluding online session. Second, a paper-based questionnaire was completed by the members of the RuralWomen. Third, the process was reflected by the designers with focus on power relations based on notes taken during the workshops. In the qualitative survey ($N = 6$ members of the RuralWomen), the advantages and disadvantages of online formats (1) in general, (2) with regard to the possibility of contributing personal views, (3) the comprehensibility of information, and (4) the level of interest and fun were collected on a Miro board. In addition, during the qualitative inquiry, the older adults also suggested discussing the advantages and disadvantages of the discussion culture at online meetings, so this has also been included. For the quantitative survey ($N = 10$) a questionnaire by Tudor et al. [28] was adapted to assess the applied methodology. In addition, information was obtained on the incentives to participate in online workshops and whether online workshops would also be of interest after the pandemic.

⁴<https://www.whatsapp.com/>

4 FINDINGS

The reflection-on-action approach revealed how the participants shaped our research, and influenced our process. It could be observed that the group was primarily driven by its most engaged individuals. However, the influence of external conditions, especially the COVID-19 pandemic, should not be neglected. The participants, the external conditions, and we as the designers influenced the participatory approach. Thus, two primary findings could be derived from our participatory process:

- (1) External factors became actors in the power relationship model,
- (2) The change of initiative altered the power relations within the model.

The power relations in this participatory process can be described by a triangle (see figure 3), which has the participants, the designers and the external conditions at one corner each. The edges of the triangle represent the relationship of these three actors. In the middle lies the decision which is based on the combination of the power relations of the actors. Through this organisation of power relations, the ratio was partly reversed when compared common patterns.

4.1 External Factors as Actors in the Relationship Model

As described in the chapter *Related Work*, external factors can be categorized into *resources*, *materials*, *unexpected events*, and *local context*. In the present study, the external factors exerted power at three distinct levels: the organizational level, the participant level, and the technological level. At the organizational level, the COVID-19 pandemic (unexpected event) necessitated a shift from offline to online participation (local context), which resulted in an increased frequency of synchronous and asynchronous communication (resources). Additionally, external factors resulted in an

Initiative Related Power

- Participant Recruiting and Invitation
- Modes of Data Assessment
- Research Questions
- Role Assignments
- Role Analysis
- Goal Definition
- Agenda Control

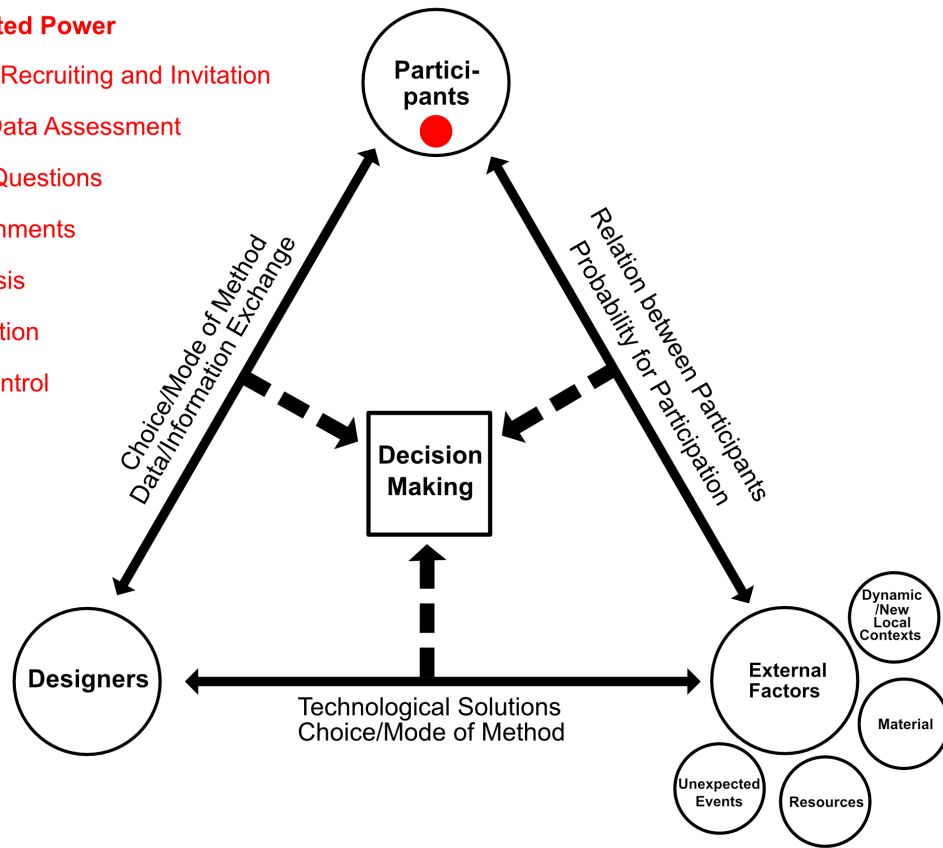


Figure 3: Illustration of the Power Relationships within the participatory process, showing that each, the participants, the researchers/developers as well as the external conditions influences decision making and thus the outcome of the process. The influence of the initiative on the power relation between participant and designer is marked in red. The aspects mentioned here are specific to our process

expansion of the participant group (local context). External participants contributed to the participatory process through supplying necessary content for the project. At the participant level, the usage of online communication tools (local context) had an influence on the discussion culture during the workshops. Participants reported feeling less confident in expressing their opinions due to difficulties in reading the mood in the digital space. Because of this, they were unsure how sensitively they had to react with their own opinions and whether their opinions were appreciated. Additionally, the technical limitations of the virtual platform made it difficult to interrupt a person directly, which further hindered the discussion culture. On a technical level, the study also found that the technology available, such as *Webex* and *Miro*, served as enabler to the participatory process. However, technical challenges became apparent during the workshops. Communication was delayed and screen contents were sometimes displayed too small. On the other hand, the participants could easily keep all the necessary documents ready and had plenty of space for them.

4.2 Altered Power Relation Between Participants and Designers

The reflection on the design process showed that in our participatory process, the power relations were different from those of the related work examined. In the following, we show the different forms in which the individual actors were able to apply their power to decision-making processes.

4.2.1 Participants. In the participatory process, participants exercised power at three distinct levels: the superordinate level, the organizational level, and the workshop level. At the superordinate level, participants were able to work independently and reflect on their actions. They also had control over the data being used and were able to provide feedback for improvement and reach consensus with the designers. At the organizational level, roles were assigned to ensure the success of the participatory development. The data assessment method (either online or postal questionnaires) was also determined, as well as the number and specific attendees for the workshop. The video conferencing platform *Webex* was recommended and used due to prior experiences. At the workshop

level, the time frame, including duration, number of meetings, and frequency, was established. Participants were also able to adjust the agenda in situ (i.e., having longer brainstorming sessions than planned). There were also intermediate sessions to discuss preliminary results. Power dynamics within the participants' group were reflected in the established and assigned roles within the participatory process (organizational team, content team, team of contemporary witnesses).

4.2.2 Designers. Designers wielded power at three distinct levels in the participatory process: the technological level, the methodological level, and the organizational level. At the technological level, designers made decisions on the methods used for data collection, established a data format for exchanging information, and controlled access to digitalized information by storing it on their own devices without easy access for participants. At the methodological level, designers had control over the selection and transparency of methods used, as well as the ability to change the agenda without consulting participants. They also held power over all aspects of evaluation, including instrument selection and analysis. Additionally, only the designers used the miro board during the workshops. Finally, at the organizational level, designers set technological requirements for roles to facilitate cooperation.

5 DISCUSSION AND FUTURE WORK

In Participatory Design, understanding the power relations and the actors involved is crucial for ensuring that all stakeholders are able to equally participate. The power relations between the three actors – users, designers and external conditions – have become more evident through the COVID-19 pandemic. In the derived triangular model (figure 3), it is mainly the strength of the relationships and influences between the nodes that shape the participatory process. In our paper we have looked at the power relations within a Participatory Design approach to software development to better understand the influencing factors and power dynamics when the initiative for participation comes from the user group itself. Based on our findings, we want to discuss three points in particular:

- (1) The change in initiative changed the power relation itself,
- (2) External conditions can act as core drivers in a participatory approach,
- (3) Long-term participation is vital for sustained Participatory Design.

5.1 Initiative Change and Power Relation

In the presented participatory project, the initiative originated from the participants, who were faced with external factors, particularly the COVID-19 pandemic and its resulting restrictions. They saw an opportunity to use Historytelling as an alternative digital solution to events that would have otherwise been cancelled or postponed. Despite these unique challenges, we found that participatory engagement remained consistent with that of traditional participatory processes. During the development phase, we observed a core group of participants who were highly engaged in the project, while others participated more loosely. Each actor, including participants, designers, and external conditions, contributed based on their own power relations, roles, and expertise within their group. This project also introduced a new form of dependency and cooperation between

participants and designers. In previous projects, designers typically formulated the research questions and solved them through a participatory approach. In the study, we had to align our questions with the interests of the participants.

As we move forward with participatory projects, it is important to consider ladders of participation (i.e., [20]) and recognize that initiative adoption is only one factor among many. In future studies, we should make a more concerted effort to involve participants in the decision-making process when it comes to selecting methods and modes of participation to promote an equal and fair participatory process.

The Power Triangle that has become evident here in the context of online collaboration should be applied and validated in additional diverse contexts and user groups. The external conditions should be identified and explicitly mentioned in the future and it should be further elaborated which conditions shape the collaboration. Questions of changing aspects over time (conditions regarding the participants and designers) and non-influenceable (external conditions) aspects have to be discussed. This raises the question of what changes over time (participants and designers aspects) and which aspects cannot be directly influenced (constant vs changing external conditions).

5.2 External Conditions as Core Drivers

In a world, where the COVID-19 pandemic has forced many organizations to shift to remote work, it has become increasingly important to understand the factors that influence the remote collaboration in participatory processes. While we found, that face-to-face meetings can be beneficial for heated discussions and interpersonal relationships, it can also be possible to hold effective meetings online. However, it is not only the *what* that matters, but also the *how* – the methods and techniques used to facilitate collaboration, regardless of whether it takes place in person or online.

When working with a group of people who do not know each other well, it may be beneficial to hold an initial in-person meeting to build relationships and establish trust before moving to remote collaboration. However, as we have seen in our research, it is possible to achieve results through remote collaboration.

In addition, the focus of the meeting may change with remote collaboration. Instead of focusing on building relationships, the focus shifts to achieving a specific goal or task. That does not mean that building relationships is not important in remote collaboration, but a different approach may be required.

The participants in our study noted that the perceived benefits of in-person meetings are often seen as drawbacks in remote meetings. However, with the right tools and techniques, it is possible to overcome these challenges and achieve effective collaboration in a remote setting.

5.3 Long-Term Participation and Sustained Participatory Design

The design and implementation of participatory processes that result in a sustainable process structure is a major challenge in HCI research. In our study, the change of initiative was a step in the right direction. However, this shift was only possible through prolonged collaboration and the development of trust and relationships

among stakeholders and designers. Despite this progress, further research is needed to explore alternative approaches to accession of power to empower the participants to an even greater extent. One solution could be to find future ways to enable participants to initiate participatory processes themselves.

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